

ATPL EASA STUDY GUIDE | 033 - FLIGHT PLANNING

Chapter 8: Fuel Monitoring, Bogey Curves & Decision Point (DPP)

Syllabus Source: CAE Oxford (pp. 291-340) | Pass Target: $\geq 90\%$ AviationExam

1. IN-FLIGHT FUEL MONITORING & BOGEY CURVES

Under EASA AIR-OPS, flight crews must record fuel remaining at each designated en-route waypoint and compare it against the operational flight plan (OFP):

- Fuel Bogey Curve (Howgozit): A graphical chart plotting Fuel Remaining versus Distance/Time. A falling curve steeper than planned indicates higher fuel consumption, headwind increase, or fuel leak!
- Minimum Fuel Declaration: 'MINIMUM FUEL' informs ATC that all planned aerodrome options have been committed to, and any unexpected delay will result in landing with less than planned final reserve fuel. (It is NOT an emergency, but priority is requested).
- Mayday Fuel Declaration: 'MAYDAY MAYDAY MAYDAY FUEL' is a distress emergency declared when estimated usable fuel on landing will be less than the 30-minute final reserve fuel. ATC will clear all conflicting traffic!

2. THE DECISION POINT PROCEDURE (DPP) / PREDETERMINED POINT (PDP)

On long-range flights, carrying 5% contingency fuel for the entire 10-hour trip creates a huge mass penalty. The Decision Point Procedure (DPP) solves this legally:

- The DPP Concept: The flight plan is split into two co-existing plans through an en-route 'Decision Point' (DP): (1) Flight to Destination via DP, with contingency fuel calculated ONLY from the Decision Point to Destination! (2) Flight to an En-Route Alternate via DP. Because contingency fuel applies only to the short leg from DP to destination, thousands of kilograms of contingency fuel are saved at take-off!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): In-Flight Fuel Checks & Decision

Standard Cockpit Procedure: Step 1: Check actual fuel on board at each primary waypoint every 30 to 60 minutes. Step 2: Compare Actual Remaining Fuel against Minimum Required Fuel on the Operational Flight Plan (OFP). Step 3: Decision Point Procedure (DPP): - At the Decision Point, if actual fuel $\geq$ fuel required from DP to Destination 1 + 20 min contingency + Alternate + Final Reserve -> PROCEED TO DESTINATION 1. - If fuel is below requirement -> DIVERT TO DESTINATION 2 (En-Route Alternate) without ever dipping into final reserve.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Fuel Savings using Decision Point Procedure (DPP)

SCENARIO (Long-Range Flight Fuel Optimization Drill): A Boeing 787 is flying from London to Tokyo: - Total Trip Fuel = 60,000 kg - A Decision Point (DP) is selected along track over Siberia - Trip Fuel from Departure to DP = 40,000 kg - Remaining Trip Fuel from DP to Tokyo Destination = 20,000 kg - Under standard fuel policy, Contingency Fuel is 5% of Total Trip Fuel - Under DPP, Contingency Fuel is 5% of Trip Fuel FROM DP TO DESTINATION! QUESTION: How much contingency fuel is saved at take-off by using the Decision Point Procedure? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Standard 5% Contingency Fuel for entire flight: - Standard Contingency = 5% of Total Trip Fuel (60,000 kg) - Standard Contingency = $0.05 \times 60,000 \text{ kg} = 3,000 \text{ kg}$. Step 2: Calculate DPP Contingency Fuel (5% from DP to Tokyo): - Trip Fuel from DP to Tokyo = 20,000 kg - DPP Contingency = 5% of 20,000 kg = $0.05 \times 20,000 \text{ kg} = 1,000 \text{ kg}$! Step 3: Calculate Fuel Savings at Take-Off: - Fuel Saved = Standard Contingency - DPP Contingency - Fuel Saved = 3,000 kg - 1,000 kg = 2,000 kg! Step 4: Operational Benefit for Dummies: - Saving 2,000 kg of contingency fuel reduces aircraft take-off mass, allowing the airline to carry 20 additional passengers or 2 tonnes of high-yield cargo, while remaining 100% compliant with EASA safety regulations! FINAL ANSWER: The Decision Point Procedure saves exactly 2,000 kg of contingency fuel at take-off.

[!] EXAM TRAP (AVIATIONEXAM) - The Decision Point Action Trap

- When reaching the Decision Point en-route: - If actual fuel remaining is AT OR ABOVE the required fuel to reach Tokyo -> CONTINUE TO DESTINATION! - If actual fuel remaining is BELOW the required fuel to reach Tokyo -> DIVERT TO EN-ROUTE ALTERNATE!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the primary operational advantage of using the DECISION POINT PROCEDURE (DPP) under EASA**

- [A]: Eliminates the need for an alternate
- [B]: It reduces the required CONTINGENCY FUEL at take-off by calculating 5% contingency only from the Decision Point to Destination, saving aircraft mass
- [C]: Allows flying without final reserve
- [D]: Permits landing in zero visibility

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under CAT.OP.MPA.180, DPP permits basing contingency fuel only on the stage from the decision point to destination (or alternate), saving substantial fuel mass at take-off.

Q2: What is the difference between declaring 'MINIMUM FUEL' and declaring 'MAYDAY MAYDAY MAYDAY FUEL'

- [A]: Minimum fuel is declared after touchdown
- [B]: 'MINIMUM FUEL' is an advisory to ATC that delay will compromise reserves (not an emergency); 'MAYDAY FUEL' is a formal distress emergency declared when estimated landing fuel is below 30-min final reserve
- [C]: There is no difference
- [D]: Minimum fuel requires immediate evacuation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Minimum fuel warns ATC that any unexpected delay will cause a fuel emergency; it does not grant priority. Mayday fuel is an emergency declaration demanding immediate priority.

Q3: On an in-flight fuel log, what does a sudden downward divergence of the actual fuel curve below the planned fuel curve indicate?

- [A]: Stronger tailwind than forecast
- [B]: An abnormal fuel burn caused by higher engine fuel consumption, lower cruise level, or a severe FUEL LEAK
- [C]: Lighter aircraft
- [D]: Cabin heating failure

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. A steep drop in the fuel curve indicates fuel is being depleted faster than planned, requiring the crew to verify engine parameters and check for fuel leaks.